

Ahmad Mustafa Anis

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EDUCATION

Carnegie Mellon University

Master of Science in AI Engineering - Electrical & Computer Engineering

Pittsburgh, PA

Dec 2027

Awards: **Block Student Fellowship on AI & Society**

International Islamic University

Bachelor of Science in Computer Science

Islamabad, Pakistan

Sep 2022

SKILLS

Libraries and Frameworks: PyTorch, Distributed Training, TensorFlow, OpenCV, FastAPI, Flask, AWS, GCP

Programming Languages: Python, C++, Spark

Databases: MySQL, PostgreSQL, MongoDB

Tools: Docker, Slurm, Spark, Weights & Biases, Qdrant, Elasticsearch, PostgreSQL, Git, Codex, Claude Code

RESEARCH EXPERIENCE & PUBLICATIONS

Research Assistant

Jun 2026 - Present

TarrLab, Carnegie Mellon University

- Developing sequential multi-objective methods for self-supervised visual representations, supervised by [Michael Tarr](#) and [Deva Ramanan](#).

AI Research Fellow

Sep 2024 - Jun 2025

Fatima Institute of Global AI Research

- Achieved **87%** accuracy on IU-Xray via **retrieval-augmented test-time inference** for Medical VLMs; advised by [Dr. Wei Peng](#).

Research Contributor

[Data Provenance Initiative](#), [MIT Media Lab](#)

Dec 2023 - Jan 2025

- Auditing large-scale LLM training data to assess web data access shifts, cultural representation, and transparency.
- S. Longpre, R. Mahari, ... **A. M. Anis** (16th Author), et al. Consent in Crisis: The Rapid Decline of the AI Data Commons. **NeurIPS 2024**
- S. Longpre, N. Singh, ... **A. M. Anis** (11th Author), et al. Bridging the Data Provenance Gap Across Text, Speech, and Video. **ICLR 2025**

Independent

- A. M. Anis**, H. Ali, M. S. Sarfraz. On the Limitations of Vision-Language Models in Understanding Image Transforms. **CVPR 2025 Workshop: Computer Vision in the Wild**.

PROFESSIONAL EXPERIENCE

Deep Learning Engineer

Mar 2024 - Aug 2026

Roll.AI - San Francisco, CA (Remote)

- Optimized **real-time video bokeh** effects by optimizing **Depth Estimation** and **Matting algorithms**, achieving **2x** performance improvement through **parallel processing** and **batch optimization** for live streaming applications.
- Developed and deployed an **Active Speaker Detection system** for multi-camera live events, automating speaker-camera associations and reducing event video editing time from 48 hours to 1 hour (**98% improvement**).
- Built a **Video Understanding pipeline** for automated chapter generation and captioning, enabling intelligent b-roll suggestions by analyzing semantic content and scene transitions across long-form video.
- Developed a **multi-view video generation pipeline** that synthesizes **novel viewpoints**, creating novel drone views for live events.
- Architected a **multimodal LLM Agent** as a centralized AI video editing system, **automating** social clip generation, multi-take video orientation and cropping, transcript enhancement, dynamic VFX and motion graphics creation to **streamline post-production** workflows.

AI Fellow

Dec 2023 - Feb 2024

PI School of AI - Rome, Italy

- 1 of 10 fellows** from 300 applicants for a [fellowship](#); developed key information extraction from long-context transcripts using LLMs.

Machine Learning Engineer

Apr 2022 - Mar 2024

Red Buffer - Islamabad, Pakistan

- Built a **fact verification** and identification pipeline with **multi-threading for a 200% speed increase**; fine-tuned LLaMAv2, Dolly-3B, GPT-Neo, and BART for negation generation to detect false news.
- Improved OCR latency from 0.6 FPS to 16 FPS** on CPU for streaming devices to blur sensitive information.
- Boosted logo recognition accuracy from **30% to 85%** during cricket matches using CLIP by OpenAI.

Deep Learning and Computer Vision Intern

Jul 2021 - Dec 2021

WortelAI - Lahore, Pakistan

- Trained a Celebrity Detection and Classification system using 3DDFAv2, ResNet50, and a 172GB dataset gaining 88% accuracy.
- Developed a real-time video search engine using CLIP by OpenAI, Qdrant for vector DB, and Elasticsearch.

ACTIVITIES & LEADERSHIP

Community Lead

Cohere Labs Community

Nov 2022 - Aug 2026

- Organized 50+** guest speaker sessions with Asian researchers; **initiated** NYU Deep Learning study group (Prof. Yann LeCun); served as Urdu Language Lead for AYA and contributed **3 Urdu datasets**.